

Abdullah Al Mahmud

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Education

PhD in Chemical and Biomolecular Engineering The Ohio State University	Columbus, OH 43210, USA 05/2025 – Ongoing
MS in Chemical Engineering Yeungnam University	Gyeongsan 38541, South Korea 03/2022 – 02/2024
BS in Chemical Engineering Bangladesh University of Engineering and Technology (BUET)	Dhaka 1000, Bangladesh 02/2016 – 02/2021

Research Interests

Computational Catalysis, DFT, MD, Solvation Thermodynamics, H₂ Production, Electrocatalysis

Journal Papers

(1 denotes co-first author)

- R Sukanya, R Karthik, **A Al Mahmud**, E Dempsey, DR Kumar, C Breslin, JJ Shim (2026). “Rational design of interface-coupled Nb-doped MoSe₂/Nb₂CT_x MXene heterostructure with enriched active sites toward enhanced hydrogen evolution reaction” (*Journal of Power Sources*) [\[link\]](#)
- I Hussain, **A Al Mahmud**, MN Lakhan, A Ali, Mohd Shkir, P. Rosaiah, Ganesh Dhakal, MZ Khan, S Rizwan, A Asghar, B Akkinepally, M Faizan (2026). “MOF-derived hollow (single-, double-, triple-, and quadruple-shell) nanostructure for energy conversion and storage applications: A review” (*Renewable and Sustainable Energy Reviews*) [\[link\]](#)
- Z Ajmal, A Qadeer, K Singh, **A Al Mahmud**, MN Lakhan, H Pradeep, B Hussain, A Ullah, HR Khan, EH Ibrahim, OJ Kewate, J Cui, P. Rosaiah, B Akkinepally, I Hussain, W Shuhang (2025). “A comprehensive review on MXenes for various applications” (*Applied Energy*) [\[link\]](#)
- W Al Zoubi, Y Sheng, MR Thalji, B Assfour, S Leoni, **A Al Mahmud**, JH Kang, AW Allaf, YG Ko (2024). “Enhanced catalytic performance via ultrasonication-plasma synergy in PtGaPCoO_x catalysts under mild conditions” (*SusMat*) [\[link\]](#)
- R Sukanya, R Karthik, **A Al Mahmud**, E Dempsey, DR Kumar, C Breslin, JJ Shim (2025). “Synergistic interface of Nb-doped MoSe₂ and NiTe heterostructure enables efficient electrocatalysis for hydrogen evolution” (*Electrochemistry Communications*) [\[link\]](#)
- SA Shah, AH Pato, S Jangra, **A Al Mahmud**, IH Sajid, P. Rosaiah, K Singh, H Bandi, MZ Khan, M Shkir, S Rizwan, TL Tamang, I Hussain (2025). “Emerging MXene/metal selenides for energy solutions: A comprehensive review” (*Nano Research*) [\[link\]](#)
- MR Thalji, A Al Mahmud, F Mahmoudi, AH Mady, DR Kumar, KF Chong, C Park, JJ Shim (2025). “Ethyl xanthate-driven in situ synthesis of Ni-Fe sulfide@Ti₃C₂T_x MXene hybrid electrodes for ultra-high-performance supercapacitors” (*Chemical Engineering Journal*) [\[link\]](#)
- I Hussain, **A Al Mahmud**, R Amna, AH Pato, U Sajjad, Z Ajmal, K Zhang (2025). “Interface and surface engineering: The nexus of MXenes, MOFs, and AI in hybrid material design for energy storage/conversion” (*Materials Today*) [\[link\]](#)
- **A Al Mahmud**¹, R Sukanya¹, R Karthik¹, DR Kumar¹, C Breslin¹, JJ Shim¹ (2024). “Synergistic promotion of active sites and enhanced electrocatalytic water splitting in metal (M = Mn, Co, Ni)-doped MoSe₂-Mo₂C heterostructures via tailored doping and interface modulation” (*Journal of Materiomics*) [\[link\]](#)
- OJ Kewate, H Jan, **A Al Mahmud**, A Batool, K Singh, P Rosaiah, M Asim, TL Tamang, I Hussain (2025). “MXene/transition metal sulfide composites in energy storage applications: A review” (*Materials Today Chemistry*) [\[link\]](#)
- I Hussain, **A Al Mahmud**, S Pandiyarajan, K Singh, EH Ibrahim, PJ Morankar, S Hussain, P Rosaiah, MZ Khan, Z Ajmal, B Akkinepally, HC Chuang, K Zhang (2025). “Practicality of MXenes: Recent trends, considerations, and future aspects in supercapacitors” (*Materials Today Physics*) [\[link\]](#)
- R Sukanya¹, R Karthik¹, **A Al Mahmud**¹, DR Kumar¹, E Kamaraj¹, C Breslin¹, JJ Shim¹ (2025). “Vanadium-doped MoSe₂ Nanosheets: Induced lattice contraction and enhanced conductivity for superior hydrogen evolution reaction” (*Electrochemistry Communications*) [\[link\]](#)
- A Milton¹, **A Al Mahmud**¹, R Sukanya¹, R Karthik¹, E Kamaraj¹, C Breslin, JJ Shim (2025). “Structural engineering of core-shell Ni₃B@Ni(BO₂)₂ on V₂M₆O₈ (0D@2D/1D) composites: advanced strategies for enhancing high energy density in asymmetric supercapacitors” (*ACS Langmuir*) [\[link\]](#)

- S Sahoo¹, A Al Mahmud¹, A Sood, G Dhakal, SK Tiwari, S Zo, HM Kim, SS Han (2025). “Microwave-assisted facile synthesis of g-C₃N₄/rGO/MoS₂ composite as the bifunctional electrocatalyst for electrochemical water splitting” (*Journal of Science: Advanced Materials & Devices*) [\[link\]](#)
- Y Sheng¹, F Hazmatulhaq¹, A Al Mahmud, MS Sayed, I Hussain, S Leoni, W Al Zoubi, YG Ko (2024). “Confinement synergy effects of supported-confined bimetal catalysts with superior stability and catalytic activity” (*Journal of Energy Chemistry*) [\[link\]](#)
- A Al Mahmud, AH Alshatteri, HS Alhasan, W Al Zoubi, KM Omer, MR Thalji, (2024). “Cu-doped Sr-MOF: Dual function active material for supercapacitor and oxygen evolution reaction” (*Electrochimica Acta*) [\[link\]](#)
- A Al Mahmud, MR Thalji, G Dhakal, Y Haldorai, WK Kim, JJ Shim (2024). “Bifunctional MoC/NiC@N-doped graphene oxide for simultaneous production of hydrogen and oxygen through overall electrochemical water splitting” (*Materials Today Nano*) [\[link\]](#)
- W Al Zoubi, A Al Mahmud, F Hazmatulhaq, MR Thalji, JH Kang, YG Ko (2024). “Origin of synergistic effects in bimetallic nanoparticles coupled with metal oxide heterostructures for accelerating catalytic performance” (*SusMat*) [\[link\]](#)
- B Kandasamy, A Al Mahmud, J Lee, I Hasan, G Palanisamy (2024). “Origin of synergistic effects in bimetallic nanoparticles coupled with metal oxide heterostructures for accelerating catalytic performance” (*Surfaces and Interfaces*) [\[link\]](#)
- SG Mohamed, MM Morad, MR Siddiqui, A Al Mahmud, W Al Zoubi (2024). “Improving the electrochemical performance of zinc sulfides by iron doping toward supercapacitor” (*Applied Materials Interfaces*) [\[link\]](#)
- HS Alhasan, AR Omran, A Al Mahmud, AH Mady, MR Thalji (2024). “Toxic Congo Red dye photodegradation employing green synthesis of zinc oxide nanoparticles using gum arabic” (*Water*) [\[link\]](#)
- DR Kumar, D Manoj, A Rosenkranz, A Al Mahmud, JJ Shim, SP Mary, S Nellaiappan (2023). “A Micro-structured growth of 1, 2, 4-triazole by electro-polymerization for highly selective detection of gallic acid and uric acid” (*Microchemical Journal*) [\[link\]](#)

Conferences, and Workshops

- “Unraveling adsorption in Hydrophilic FAU zeolite: Coupling DFT-derived charge distributions with statistical mechanics simulations” (*Midwest Thermodynamics and Statistical Mechanics Conference (MTSM 2026)*, University of Minnesota, Twin Cities), [Oral Presentation on 06/16/2026]
- “Polypyrrole-assisted preparation of γ -MoC/NiC composite with reduced graphene oxide for overall electrochemical water splitting” (*International Conference on Advanced Materials & Sustainable Energy Technologies-2023 (AMSET)*, Sunway University, Malaysia), [Oral Presentation on 10/30/2023]

Work Experience

Graduate Research Assistant (GRA)

[Getman Fluid/Solid Interface Engineering Group](#), The Ohio State University

(05/2025 – Ongoing)

Computational Catalysis

- Contributions of rotational flexibility of adsorbates in solvation thermodynamics of BEA and FAU zeolites
- Improvement of solvation thermodynamics in microporous zeolite networks for solvent-mediated interactions using multi-scale modeling and quantum calculations
- Use of first principles and machine learning approaches to design suitable catalysts for hydrogen production during catalysis, or electrocatalysis

Graduate Research Assistant (GRA)

[Nano Energy Materials and Processes Laboratory](#), Yeungnam University

(03/2022 – 03/2024)

Electrochemical Water Splitting

- Transitional multi-metallic carbide composites with graphene oxide using polymerization and carbonization
- Copper-doped strontium metal-organic framework for oxygen evolution reaction and energy supercapacitor applications
- Bimetallic nanoparticles coupled with metal oxide using ultrasound-assisted low-voltage plasma electrolysis

Energy Storage

- Bimetallic sulfide composite with MXene (Ti₃C₂T_x) for high-performance supercapacitor applications
- Bimetallic oxide composite with a metal boride nanoparticle for high-performance asymmetric supercapacitor applications

Undergraduate Research

Part-time research assistant, BET Group (2019-2020), Department of Chemical Engineering, BUET

Quantification of industrial wastewater and antibacterial paper development

- Quantitative analysis of different parameters for the wastewater of textile and pharmaceutical industries

Graduate Teaching Assistant (GTA)

School of Chemical Engineering, Yeungnam University

Theoretical course – Chemical Engineering Thermodynamics I (Spring 2022 & 2023), Fluid Mechanics (Fall 2022)

- Scheduled weekly activities, proctored in the exams, graded the homework, quizzes, and assignments

Laboratory Course – Chemical Engineering Thermodynamics Lab (lab coordinator, Fall 2022)

- Taught and engaged with undergraduate students' participation in experiments and research projects

Skills

• Material Characterization	XRD, FESEM, TEM, HRTEM, EDS, XPS, EPR, Raman, FTIR, UV-Vis
• Characterization Analysis	ImageJ, Digital Micrograph, CasaXPS, XPSpeak41, X'Pert HighScore
• Electrochemical Analysis	CV, EIS, LSV, GCD, chrono-amperometry, chrono-potentiometry, battery cycler
• Material synthesis	Solvothermal, Etching, Calcination, Polymerization, Carbonization
• Molecular Visualization	Avogadro, VESTA, Jmol, OVITO, GaussView, VMD
• Computational Analysis	CP2K, Gaussian, LAMMPS, GROMACS, VASP, MobaXterm
• Programming Software	Python, R
• Drawing Tools	Blender (3D), Photoshop, AutoCAD (2D), MS Visio
• Writing & Illustrations	LaTeX, MS Word, Mendeley Reference Manager, EndNote, Zotero
• Database and Spreadsheet	Origin, MS Excel, Kobo Toolbox
• Presentation	MS PowerPoint

Achievements & Awards

- Best Poster Award, 22nd National Symposium of Clean Technology (05/2023)
- BK21 scholarship supported by Brain Korea21 (four) Project for graduate research (03/2022 – 02/2024)
- International student scholarship for MS, Yeungnam University (03/2022 – 02/2024)
- Technical Scholarship (undergraduate), BUET (02/2016 – 02/2020)

Affiliations

- Member, Korean Research Institute, Researcher ID – 12933241
- Graduate Student Member, American Institute of Chemical Engineers (AIChE), ID – 009902469197
- Member, Institute of Clean Technology, School of Chemical Engineering, Yeungnam University

Internships & Training

- Undergraduate Internship for two weeks at *Eastern Refinery Limited, Chattogram, Bangladesh* (04/2019)
- Full day training on Sulfuric Acid production at *Wata Chemicals Limited, Narayanganj, Bangladesh* (05/2018)

Extracurricular Activities

- Treasurer, ExCom 2025-26, Bangladesh Graduate Student Association (BGSA), The Ohio State University (10/2025 – Present)
- Student Volunteer, 21st Symposium of Clean Technology, Yeungnam University (05/2022)
- Student Volunteer, 21st International Symposium of Clean Technology, Yeungnam University (11/2022)

References

- Rachel B. Getman** (PhD Advisor)
Professor & Bernice L. Claugus Endowed Chair
William G. Lowrie Department of Chemical and Biomolecular Engineering
The Ohio State University, USA
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- Jae-Jin Shim** (MS Advisor)
Emeritus Professor
Department of Chemical Engineering
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- Raj Karthik** (Research Collaborator)
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